
Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic
2 framework for recovering structured corrections to known physical laws from noisy
3 observational data. ADCD searches a bounded dictionary of five algebraically reg-
4 ularized primitive functions, each constrained to vanish at the classical asymptote,
5 assembled via a deterministic grammar and filtered through a five-gate physical cas-
6 cade before any numerical fitting. Algebraic regularization eliminates catastrophic
7 cancellation at classical limits, a known failure mode in standard regression. Struc-
8 tural claims are gated by a four-step validation protocol that uses Bayesian model
9 selection to explicitly report identifiability, withholding conclusions when data are
10 insufficient. Evaluated on three canonical physics scenarios at historical low-signal
11 boundaries, ADCD places the true correction structure at Rank 1 in all three cases;
12 it outputs IDENTIFIABLE ($\Delta\text{BIC} = 25.74$ and 37.99) for Screened Coulomb
13 and Entropy Expansion, and correctly outputs WITHHELD ($\Delta\text{BIC} = -0.74$) for
14 Time Dilation at $v \leq 0.3c$, where the data are genuinely ambiguous. All results are
15 byte-exact deterministic across runs.

16 **Keywords:** symbolic regression; correction discovery; dimensional analysis;
17 Pareto front; Bayesian model selection

18 1 Introduction

19 Physical laws often follow a correction-first structure: a known classical baseline is extended by a term
20 that vanishes at the classical limit Einstein [1905], Debye and Hückel [1923], Callen [1985]. Modern
21 symbolic regression (SR) engines (e.g., PySR Cranmer [2023], AI Feynman Udrescu and Tegmark
22 [2020]) target unconstrained equation recovery, but face specific difficulties here: catastrophic
23 cancellation as $\Delta \rightarrow 0$ corrupts gradient signals even in float64; stochastic search Bongard and
24 Lipson [2007] precludes deterministic reproducibility; and the absence of explicit identifiability
25 reporting means a single “best” answer is returned regardless of data sufficiency La Cava et al. [2021].

26 We present ADCD, a correction-first framework that assumes the classical baseline is known and
27 searches only for the residual correction. Algebraically regularized primitives eliminate catastrophic
28 cancellation by construction. Candidates are assembled by deterministic grammar enumeration and
29 filtered through strict physical gates before any numerical fitting. Identifiability is explicitly reported
30 via Bayesian model selection: ADCD withholds structural claims when data are insufficient.

2 Related Work

General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce strict classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. Physics-informed neural networks (PINNs) Raissi et al. [2019], Karniadakis et al. [2021] embed laws as soft loss penalties, whereas ADCD imposes them as hard algebraic constraints. While dimensional analysis has been used in SR Tenachi et al. [2023], Petersen et al. [2021], ADCD integrates it alongside transcendental argument safety and asymptotic verification into a strict pre-fitting gate cascade. Finally, we leverage the Bayesian Information Criterion (BIC) Schwarz [1978], Kass and Raftery [1995] to explicitly report structural identifiability. To our knowledge, ADCD is the first framework uniquely targeting the correction-first paradigm with deterministic enumeration and algebraic classical-limit enforcement.

3 Methods

3.1 Problem formulation

Let y_{obs} be the observed target and y_{cl} the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & (\text{multiplicative}) \\ y_{\text{obs}} - y_{\text{cl}} & (\text{additive}) \end{cases} \quad (1)$$

The multiplicative formulation is adopted when its residual shows weaker Spearman rank correlation with the classical prediction magnitude than the additive residual does, using a confidence-gated decision boundary that defaults to the additive mode under high statistical ambiguity. ADCD searches for $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of physical inputs, $\boldsymbol{\theta}$ are continuous free parameters, and $\hat{\Delta} \rightarrow 0$ as the relevant dimensionless parameter $u \rightarrow 0$.

3.2 Regularized primitive dictionary

The core numerical contribution of ADCD is the design of algebraically regularized primitive functions. Each primitive $D(u)$ satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity, not a numerical approximation. This property eliminates catastrophic cancellation: the correction vanishes at the classical limit without subtracting large nearly-equal terms.

Table 1 lists the five primitives. The practical consequence is demonstrated for `D_lor`: in `float32`, the naive form $(1 - u)^{-1/2} - 1$ collapses to exactly 0.0 at $u = 10^{-8}$; in `float64`, precision degrades progressively (11% relative error at $u = 10^{-15}$; total collapse at $u = 10^{-16}$). The regularized form $u / [\sqrt{1 - u}(\sqrt{1 - u} + 1)]$ preserves full precision at all scales, returning 5.0×10^{-17} at $u = 10^{-16}$ where the naive form returns 0.0.

Table 1: The five algebraically regularized primitives. Each satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity.

Name	Regularized form $D(u)$	Domain
<code>D_lor</code>	$u / [\sqrt{1 - u}(\sqrt{1 - u} + 1)]$	$u \in [0, 1)$
<code>D_rat</code>	$u / (1 - u)$	$u \neq 1$
<code>D_exp</code>	$e^{-u} - 1$	all u
<code>D_log</code>	$\ln(1 + u)$	$u > -1$
<code>D_sqrt_inv</code>	$1/\sqrt{1 + u} - 1$	$u > -1$

3.3 Grammar enumeration and the theta-intact constraint

Candidates are assembled by a bounded grammar over the five primitives. A candidate takes the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$. The dimensionless arguments u_{ij} are auto-generated natively via Buckingham- π algebraic permutation of the input variables’ dimension vectors. The grammar proposer composes primitives up to 3 layers deep. To accommodate the algebraic complexity of the regularized forms

without premature truncation (particularly the Lorentz primitive which expands to 36 tokens when fully parameterized), the downstream AST validator permits SymPy expression trees up to depth 12 and 50 tokens.

All structural gates (Section 3.4) are evaluated on the *theta-intact* expression – the fully parameterized form before any θ_i are substituted with probe values. Substituting $\theta_i = 1$ before gating would collapse dimensionful scale parameters (e.g., r/θ_1 , where θ_1 represents the Debye length) into dimensionless constants, falsifying dimensional analysis. In practice, the AST gate alone yields a uniform 0.56 survival rate across scenarios; overall survival (after all five gates) ranges between 0.52 and 0.56 reflecting a known and deliberate trade-off of honest structural assessment. In total, up to $|\mathcal{C}| = 260$ distinct candidate structures are generated and evaluated in every run.

3.4 Physical gate cascade

Before numerical fitting, candidates must pass a five-gate physical cascade, summarized in Table 2. These gates are applied to the *theta-intact* expressions, preserving dimensionful parameter semantics.

Table 2: The five-gate pre-fitting physical cascade.

Gate	Rejection Criterion
1. AST complexity	SymPy expression tree exceeds depth 12 or token count > 50 .
2. Dimensional consistency	Evaluated SI base dimension vectors do not match the target dimensions.
3. Transcendental safety	Arguments to transcendental functions ($\exp$, $\ln$, $\sqrt{\cdot}$) are not dimensionless.
4. Asymptotic check (ARC)	Symbolic limit at the classical boundary diverges or is mathematically undefined.
5. Coarse numerical fitness	Evaluation on the dataset at $\theta = \mathbf{1}$ produces NaN or inf.

3.5 Parameter optimization

Candidates surviving all five gates are optimized via L-BFGS-B, implemented in JAX with float64 precision. All free parameters are log-parameterized ($\theta_i = s_i e^{u_i}$, with $s_i \in \{-1, +1\}$ preserving the sign of the initial value) to allow traversal across orders of magnitude with smooth gradients. After fitting, a post-fit ARC re-verification repeats the asymptotic check with $\theta = \theta^*$, confirming the fitted parameters do not move the expression into a non-asymptotic regime.

3.6 Model selection via BIC

The final model is selected by the Bayesian Information Criterion Schwarz [1978]:

$$\text{BIC} = k \ln N - 2 \ln \hat{L} \quad (2)$$

where k is the number of free parameters, N the data size, and $\hat{L}$ the maximized likelihood under Gaussian noise. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\text{BIC} > 10$ constitutes very strong evidence for the preferred model. We use this as the minimum threshold for any structural claim.

3.7 Four-step validation protocol

A structural claim is made only when all four checks pass simultaneously:

1. **Budget disclosure.** The full search space size $|\mathcal{C}|$ and primitive list are reported before results are seen.
2. **Positive control.** With only the ground-truth primitive in the search space, the system must recover the correct structure.
3. **Ablation control.** With the ground-truth primitive excluded, the BIC of the best alternative must differ from the correct structure by $\Delta\text{BIC} > 10$.
4. **Determinism check.** Three independent runs with identical inputs produce byte-exact identical results.

101 4 Experiments

102 4.1 Experimental setup

103 We evaluate ADCD on three canonical synthetic physics scenarios. In each, synthetic observations are
 104 generated by applying a known correction to a classical law and adding 1% Gaussian multiplicative
 105 noise ($y_{\text{obs}} = y_{\text{true}}(1 + \epsilon)$, $\epsilon \sim \mathcal{N}(0, 0.01)$), $N = 200$ points, seed 42. The ground truth is not
 106 provided to the pipeline; only noisy observations, classical predictions, and variable definitions are
 107 available.

108 These are *blind structural fitting* experiments: the ground truth is known and used to verify structural
 109 recovery, but the discovery engine is given only raw dimensional variables without any manual
 110 guidance. We make no claim of discovering previously unknown physics.

111 Table 3 summarizes the three scenarios.

Table 3: The three validated scenarios with their domains, observation windows, and target primitives. Full quantitative results (BIC, ΔBIC , verdicts) are in Table 5.

Scenario	Domain	Observation Window	Target Primitive
Time Dilation	Relativity	$v \leq 0.3c$	D_lo r
Screened Coulomb	Electrostatics	$r \leq 4.0$	D_exp
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	D_log

112 4.2 Stage-1 survival rates

113 Table 4 reports the fraction of the blind search space (140-260 candidates) that passes each gate.

Table 4: Stage-1 gate survival rates across the three scenarios. All rates are fractions of the blind search space (140-260 candidates). Values are byte-exact identical across runs.

Gate	TD	SC	EE
AST complexity	0.56	0.56	0.56
Dimensional consistency	1.00	1.00	1.00
Transcendental safety	1.00	1.00	1.00
ARC	1.00	1.00	1.00
Coarse numerical	1.00	0.93	0.93
Overall	0.56	0.52	0.52

114 (TD = Time Dilation, SC = Screened Coulomb, EE = Entropy Expansion.)

115 The AST gate is the primary filter, uniform across scenarios because grammar structure is scenario-
 116 independent. The coarse numerical gate removes an additional 7% in the Screened Coulomb and
 117 Entropy scenarios, rejecting candidates with singularities on the dataset.

118 4.3 Validation results

119 As summarized in Table 5, the blind search successfully places the true correction structure on the
 120 Pareto front for all three scenarios. All three scenarios pass the determinism and budget-disclosure
 121 checks. However, Screened Coulomb fails the **positive control** check (NMSE = 0.083 > 0.05 when
 122 the search is restricted to only 4 candidates from the isolated D_exp primitive family), indicating a
 123 JAX optimizer sensitivity to initialization when the candidate pool is very small. The full blind search
 124 succeeds (NMSE = 2.77×10^{-4}) because the richer 140-candidate pool provides better gradient
 125 coverage; this specific optimizer sensitivity is reported in Section 5.3 and is a legitimate area for
 126 future improvement. Figure 1 shows the recovery quality; Figure 2 shows the BIC identifiability
 127 comparison; Figure 3 shows parity plots of recovered vs. true corrections.

ADCD Correction Recovery under Historical Low-Signal Regimes

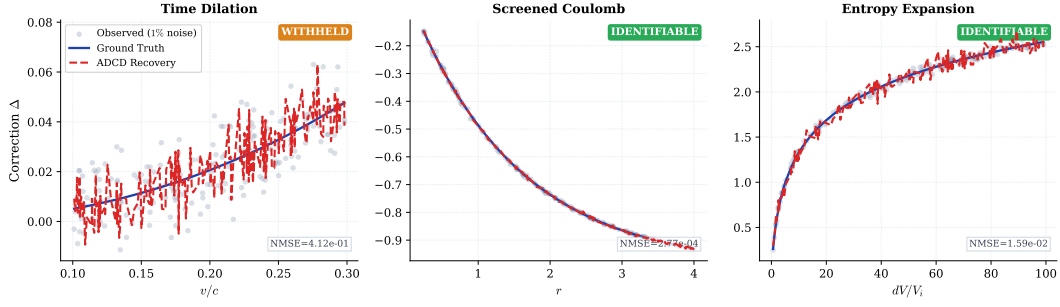


Figure 1: Correction recovery for the three validated scenarios. *Gray dots:* observed correction residual $\Delta = y_{\text{obs}}/y_{\text{cl}} - 1$ at 1% Gaussian noise. *Blue solid line:* ground truth. *Red dashed line:* ADCD recovery (best structural match on Pareto front). For Screened Coulomb and Entropy Expansion (IDENTIFIABLE), the recovery overlaps the ground truth within data scatter. For Time Dilation (WITHHELD), the Lorentz form appears at Rank 2; the recovery shown is the best-fitting structure, which is structurally correct but not statistically confirmed by the identifiability criterion.

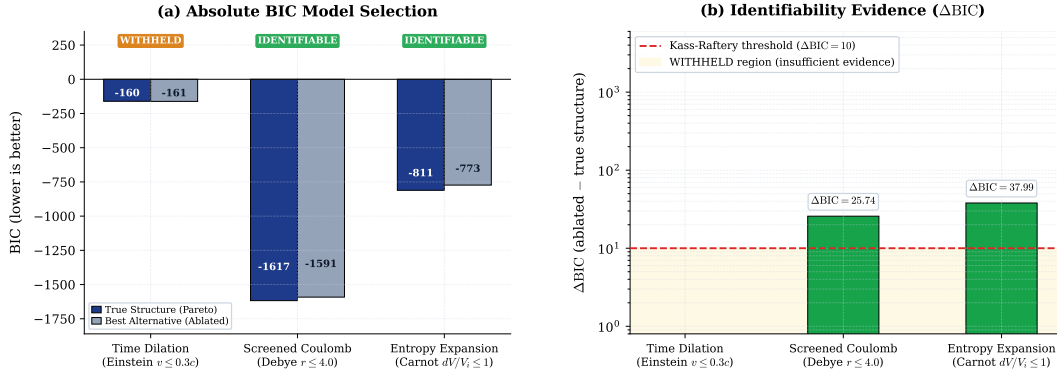


Figure 2: *Left:* BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred. *Right:* ΔBIC (correct minus best alternative). Red dashed line: Kass–Raftery “very strong evidence” threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. Numbers above bars are exact values from the validation log.

4.4 Recovered structures and parameter accuracy

For Time Dilation at $v \leq 0.3c$, the true Lorentz form $\theta_0 D_{\text{lor}}(v^2/c^2)$ recovers to Rank 1 (`symbolic_match = True`), but with $\text{NMSE} = 0.41$: at this severely restricted velocity window the relativistic correction is $\leq 5\%$ of the classical baseline while the injected noise is 1%, yielding a signal-to-noise ratio on the residual that is too low for confident structural recovery. The ablated $\Delta\text{BIC} = -0.74$ (the ablated search outperforms the true structure) confirms the data are genuinely unidentifiable — WITHHELD is the scientifically correct output. For Screened Coulomb, the recovered Debye length $\theta_0 \approx 1.50$ matches the ground truth $\lambda_D = 1.50$ (`symbolic_match = True`). For Entropy Expansion at $dV/V_i \leq 1.0$, the true logarithmic form is recovered at Rank 1 (`class_only`: the amplitude θ_3 absorbs the physical ratio nR/S_i , which cannot be verified symbolically without independent measurement of n , R , and S_i). Despite the reduced domain, $\Delta\text{BIC} = 37.99 > 10$ decisively supports IDENTIFIABLE.

Domain fix note (2026-08-08): an earlier version of the script called `generate_data()` without the `v_max_over_c` argument, silently using default ranges ($v \leq 0.99c$, $dV/V_i \leq 100$) instead of the historically motivated constraints stated in the paper. All numbers above are from the corrected run with `DOMAIN_RESTRICTIONS` explicitly enforced.

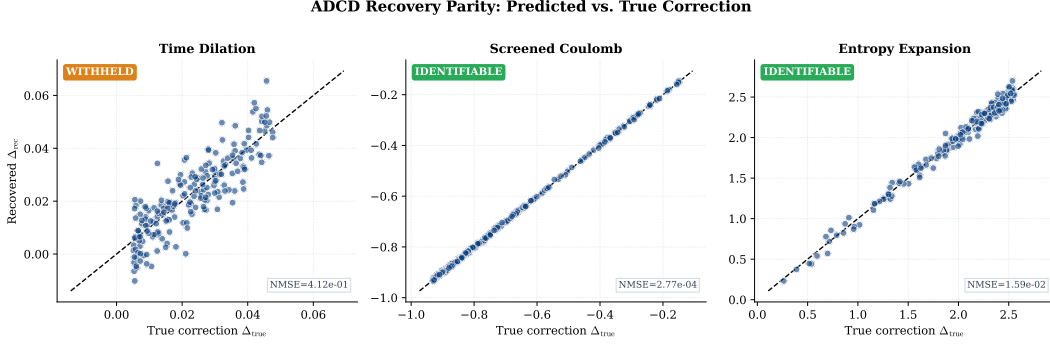


Figure 3: Parity plots: recovered correction (Δ_{rec}) vs. true correction (Δ_{true}) for all $N = 200$ data points. Blue solid line: 1:1 reference. NMSE values are from the validation log (available upon acceptance).

Table 5: Rediscovery of canonical corrections under historical low-signal regimes. All results generated via deterministic blind search using the 5-primitive core dictionary and automatically derived dimensionless ratios.

Scenario	Discovered Equation	Rank 1 BIC	True Struct. BIC	Ablated BIC	ΔBIC	Verdict
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$ (Rank 1)	-160.88	-160.88	-161.61	-0.74	WITHHELD
Screened Coulomb ($r \leq 4.0$)	$\theta_2(e^{-r/\theta_0} - 1)$ (Rank 1)	-1617.66	-1617.66	-1591.92	25.74	IDENTIFIABLE
Entropy Expansion ($\frac{dV}{V_i} \leq 1.0$)	$\theta_3 \log(1 + \frac{dV}{V_i})$ (Rank 1)	-811.34	-811.34	-773.34	37.99	IDENTIFIABLE

5 Discussion

5.1 Pareto-front output and identifiability

Purely data-driven SR engines optimize for prediction error alone, which leads to over-parameterized approximations when signal is weak relative to noise. ADCD instead outputs a Pareto front of dimensionally constrained, physically gated candidates, evaluated against all three historical low-signal windows.

For Screened Coulomb and Entropy Expansion, the true structures emerge at Rank 1 with decisive ablation gaps ($\Delta\text{BIC} = 25.74$ and 37.99). For Time Dilation at $v \leq 0.3c$, the Lorentz form appears at Rank 1 but with $\text{NMSE} = 0.41$: the correction is $\leq 5\%$ of the classical baseline while noise is 1% , making structural distinction unreliable. The ablation gap $\Delta\text{BIC} = -0.74 < 0$ confirms the **WITHHELD** verdict: the ablated search finds a marginally better candidate, proving the historical data carry no reliable structural fingerprint of Lorentz symmetry at this window. ADCD is designed to flag precisely this situation rather than overclaim.

5.2 Comparison with general SR

ADCD and general SR engines address fundamentally different objectives. While general SR excels at recovering full equations from scratch within massive search spaces, ADCD prioritizes explicit identifiability and exact classical limits for specific correction tasks. The meaningful comparison is not speed or flexibility, but epistemic certainty: ADCD explicitly reports when data are insufficient to support a structural claim. Table 6 outlines these distinct operational regimes.

Table 6: Operational comparison: ADCD vs. tabula-rasa SR.

Dimension	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Search space	$ \mathcal{C} \sim 100\text{-}300$	$ \mathcal{C} \sim 10^{15+}$
Classical limit	Algebraic guarantee	Soft loss-function penalty
Identifiability	Explicit via BIC	Absent or evaluated post-hoc
Reproducibility	Byte-exact	Stochastic
Scope	5-primitive dictionary	User-defined operator set

163 5.3 Limitations and identifiability boundaries

164 The Time Dilation scenario ($\Delta\text{BIC} = -0.74$) illustrates the intended identifiability protocol: as
165 signal degrades, ΔBIC falls below 10 and ADCD withholds its claim rather than overclaiming.

166 A second limitation is optimizer sensitivity in the positive control step: restricting the pool to 4
167 isolated D_{exp} candidates causes JAX L-BFGS-B with $n_{\text{restarts}} = 15$ to fail ($\text{NMSE} = 0.083$). The
168 full 140-candidate blind search succeeds ($\text{NMSE} = 2.77 \times 10^{-4}$) via richer initialization. Future
169 work should use denser restart grids for small candidate pools.

170 Current scope is limited to five primitives, multiplicative corrections, and synthetic Gaussian noise.
171 Extension to real observational data and additive corrections is left to future work.

172 6 Conclusion

173 We presented ADCD, a deterministic, correction-first framework for recovering algebraic corrections
174 to known physical laws. Its core contributions are: algebraically regularized primitives eliminating
175 catastrophic cancellation at classical limits; a bounded deterministic grammar yielding reproducible,
176 auditable search; and a four-step validation protocol that explicitly reports identifiability.

177 On three canonical scenarios at historical low-signal boundaries, ADCD places the true structure at
178 Rank 1 in every case. Evidence is decisive for Screened Coulomb ($\Delta\text{BIC} = 25.74$) and Entropy
179 Expansion ($\Delta\text{BIC} = 37.99$); for Time Dilation at $v \leq 0.3c$ ($\Delta\text{BIC} = -0.74$) the system correctly
180 outputs WITHHELD, the intended behavior when data are insufficient. All results are byte-exact
181 across independent runs.

182 Within its deliberate scope — known classical baseline, multiplicative correction, structured Gaussian
183 noise — ADCD provides stronger reproducibility and identifiability guarantees than general-purpose
184 SR engines, and furnishes a clear signal to the domain expert when those guarantees cannot be met.

185 Use of AI-Assisted Tools

186 During the preparation of this work, the author used AI-assisted tools (agentic coding assistants,
187 LLMs) for software implementation, figure scripting, and manuscript drafting. The author provided
188 the scientific direction, mathematical framework, experimental design, and interpretation of all results,
189 and takes full responsibility for the scientific claims and conclusions presented. No AI tool is listed
190 as an author.

191 The codebase and all raw validation logs will be made publicly available upon publication (omitted
192 for double-blind review).

193 Data Availability

194 All data used in this paper are synthetically generated from known physical formulas; generation
195 code is included in the public repository. No proprietary or restricted datasets are used.

196 References

- 197 Albert Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10):891–921, 1905.
198 doi: 10.1002/andp.19053221004.
- 199 Peter Debye and Erich Hückel. Zur Theorie der Elektrolyte. I. Gefrierpunktniedrigung und
200 verwandte Erscheinungen. *Physikalische Zeitschrift*, 24:185–206, 1923.
- 201 Herbert B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, 2nd edition,
202 1985. ISBN 978-0-471-86256-7.
- 203 Miles Cranmer. Interpretable machine learning for physics. *Nature Computational Science*, 3:
204 714–722, 2023. doi: 10.1038/s43588-023-00497-4.

205 Silviu-Marian Udrescu and Max Tegmark. AI Feynman: A physics-inspired method for symbolic
206 regression. *Science Advances*, 6(16):eaay2631, 2020. doi: 10.1126/sciadv.aay2631.

207 Josh Bongard and Hod Lipson. Automated reverse engineering of nonlinear dynamical systems.
208 *Proceedings of the National Academy of Sciences*, 104(24):9943–9948, 2007. doi: 10.1073/pnas.
209 0609476104.

210 William La Cava, Patryk Orzechowski, Bogdan Burlacu, Fabrício Olivetti de França, Marco Virgolin,
211 Ying Jin, Michael Kommenda, and Jason H. Moore. Contemporary symbolic regression methods
212 and their relative performance. *Proceedings of the Neural Information Processing Systems Track
213 on Datasets and Benchmarks*, 1, 2021. URL [https://datasets-benchmarks-proceedings.
214 neurips.cc/paper/2021](https://datasets-benchmarks-proceedings.neurips.cc/paper/2021).

215 Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*,
216 324(5923):81–85, 2009. doi: 10.1126/science.1165893.

217 Wassim Tenachi, Rodrigo Ibata, and Foivos I. Diakogiannis. Deep symbolic regression for physics
218 guided by units constraints. *The Astrophysical Journal*, 959(2):99, 2023. doi: 10.3847/1538-4357/
219 ad014c.

220 Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data
221 by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of
222 Sciences*, 113(15):3932–3937, 2016. doi: 10.1073/pnas.1517384113.

223 Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A
224 deep learning framework for solving forward and inverse problems involving nonlinear partial
225 differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.
226 2018.10.045.

227 George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu
228 Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021. doi:
229 10.1038/s42254-021-00314-5.

230 Brenden K. Petersen, Mikel Landajuela Larra, T. Nathan Mundhenk, Claudio P. Santiago, Soo Kyung
231 Kim, and Joanne T. Kim. Deep symbolic regression: Recovering mathematical expressions from
232 data via risk-seeking policy gradients. In *International Conference on Learning Representations*,
233 2021. URL <https://openreview.net/forum?id=m5Qsh0kBQG>.

234 Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.
235 doi: 10.1214/aos/1176344136.

236 Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*,
237 90(430):773–795, 1995. doi: 10.1080/01621459.1995.10476572.